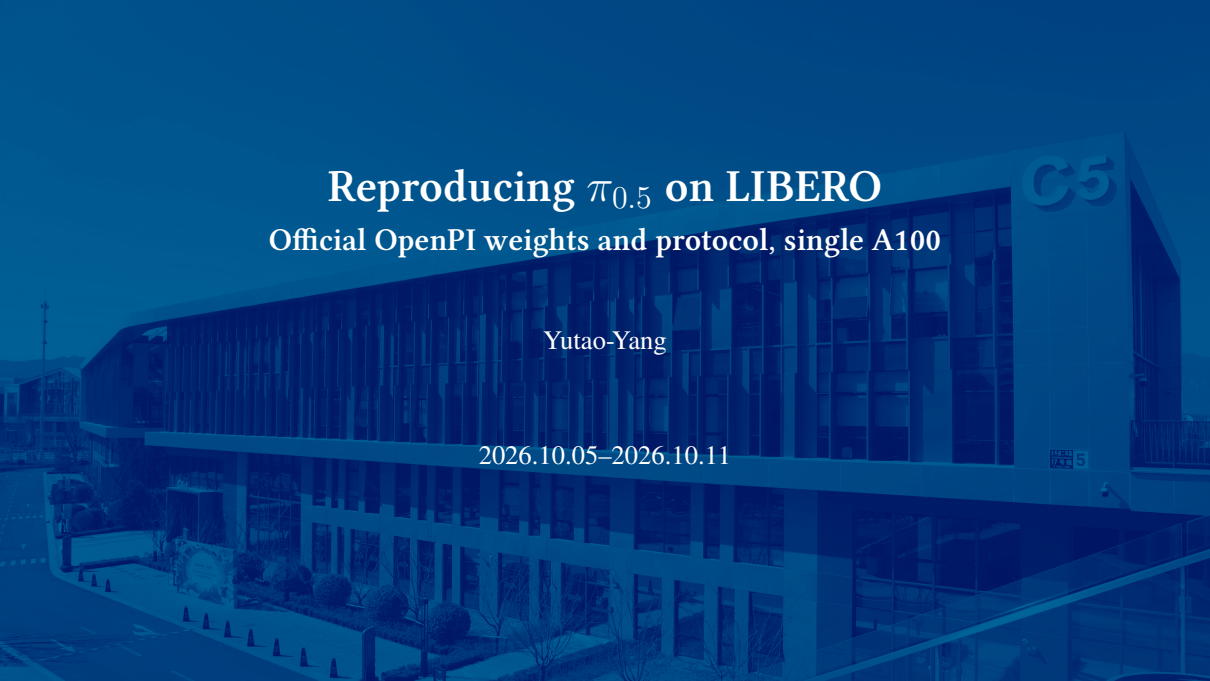


Reproducing $\pi_{0.5}$ on LIBERO

Official OpenPI weights and protocol, single A100

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2026.10.05–2026.10.11

Contents

Motivation

Benchmark and setup

Result

Compute

Teaching cases

Normal failures

Takeaway





Motivation

Why reproduce $\pi_{0.5}$?

- $\pi_{0.5}$ is a vision–language–action model: two camera streams and a language task in, a continuous 10×7 action chunk out.
- The authors report **96.85 %** over the four standard LIBERO suites.
- Question: does the *released* checkpoint, driven by the *official* OpenPI implementation, reproduce that number on one A100?

What is held fixed

Released LIBERO weights, authors' task protocol, environment and budgets. No retraining, no per-seed selection. A P10 highest-precision run is kept as a *control*, not as the main table.



Benchmark and setup

- 40 tasks $\times$ 50 fixed initial states = **2000** closed-loop episodes.
- 20 Hz control; 256 RGB frames, 180° rotation, padded to 224.
- Success is the official `done/goal` signal; timeouts count as failures.

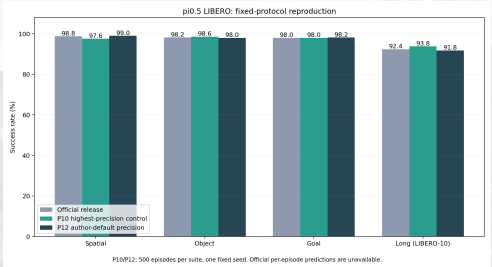
Suite	Focus	Official
Spatial	spatial relations	98.8 %
Object	target objects	98.2 %
Goal	goal states	98.0 %
LIBERO-10	longer horizon	92.4 %
Mean	40 tasks	96.85 %

Official numbers come from the OpenPI release table, not from this run.



Result

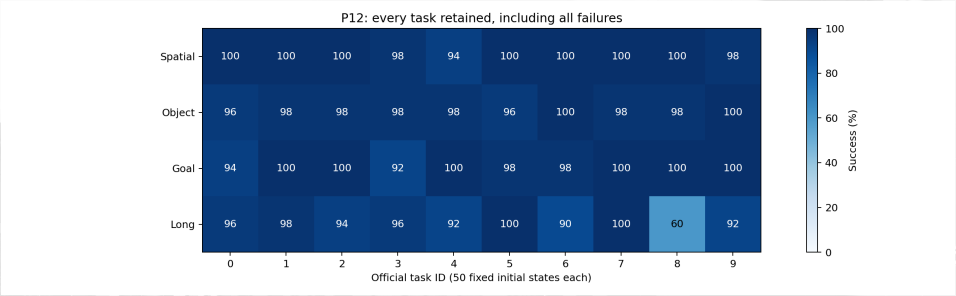
- Reproduced: $\mathbf{1935/2000 = 96.75}\,\%$.
- Official: $96.85\,\%$ (-0.10 percentage points).
- All 2000 episodes retained; all 64 761 inferences and 320 142 executed steps finite.



Per suite (success / 500)

Spatial 495 Object 490 Goal 491 LIBERO-10 459. P10 precision control: $97.00\,\%$.

Every task


P12: every task retained, including all failures
Spatial
Object
Goal
Long
100 100 100 98 94 100 100 100 100 98
96 98 98 98 98 96 100 98 98 100
94 100 100 92 100 98 98 100 100 100
96 98 94 96 92 100 90 100 60 92
0 1 2 3 4 5 6 7 8 9
Official task ID (50 fixed initial states each)
Success (%)
100
80
60
40
20
0



Compute

- End-to-end wall time: 9673.44 s (≈ 2 h 41 m) — the four suites overlap, so they must not be summed.
- 64 761 model queries; 320 142 executed control steps.

Suite	Wall	Steps
spatial	85.36 min	52 625
object	95.89 min	70 938
goal	85.43 min	57 342
LIBERO-10	147.45 min	139 237

Predict 10 actions, execute the first 5, then replan.



Teaching cases

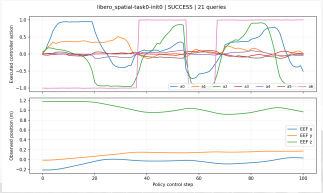
Genuine model inputs: external and wrist views



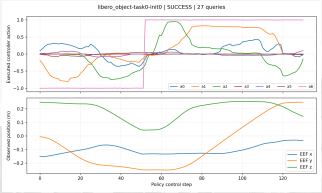
spatial object goal i0 goal i1

First real query of four teaching cases. These six P9 cases are engineering cases and do not replace the 2000-episode score.

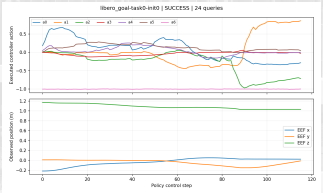
Executed actions



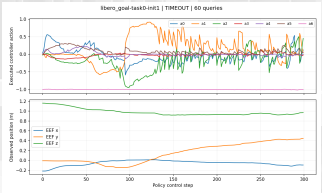
spatial t0 i0 (success, 101 steps)



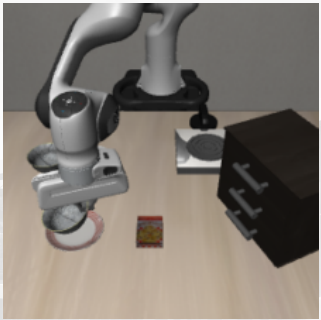
object t0 i0 (success, 131 steps)



goal t0 i0 (success, 116 steps)



goal t0 i1 (timeout failure, 300 steps)



Play video

spatial t0 i0 — success



Play video

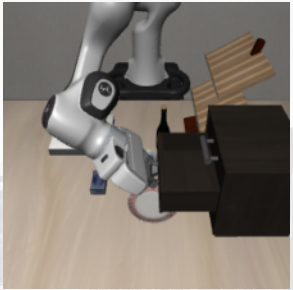
goal t0 i1 — timeout failure

Last query of each run. Videos are written at 10 fps for viewing, so they last about twice the simulated action time. Click a button to open the file next to the PDF.



Normal failures

An open drawer is not success



“open the middle drawer” — **success**

middle drawer -0.14426769



“open the middle drawer” — **failure**

middle drawer 0.00147690

bottom drawer opened

The BDDL goal requires the *middle*-drawer joint below -0.14. Both rolls were replayed on CPU with a maximum state error of 0 and agreed `done/goal` signals.



Takeaway

What this shows --- and what it does not

- The released checkpoint **reproduces the official score** to within 0.10 percentage points using the official implementation.
- Agreement is about the aggregate number: the release publishes no per-episode predictions, so per-sample identity is not claimed.
- The P12/P10 gap is a single-run precision control and is *not* a stable precision advantage.
- Non-finite engineering failures (NaN GPU inputs) are reported separately from genuine task failures such as the drawer case.
- In this configuration the 8-D state is *not* a network condition, so no state-conditioned claim is made.
- This is a baseline reproduction, not a fair architecture comparison with the other model in this project.

- $\pi_{0.5}$: Black et al., *A Vision–Language–Action Model with Open-World Generalization*, arXiv:2504.16054.
- OpenPI reference implementation: revision `215abfb2`.
- LIBERO: Liu et al., *Benchmarking Knowledge Transfer for Lifelong Robot Learning*, arXiv:2306.03310.

Full report: `reports/2026.10.05-2026.10.11-pi05-libero/`